CLINICAL EVIDENCE REVIEW · V2.0 · SUPERSEDES V1

Crohn's disease and investigational retatrutide 4 mg

What the phase 3 data now say about this dose, what the IBD evidence does and does not support, and how to tell a flare from a side effect.

CASE PROFILE	Woman, 40, ileal Crohn's disease with chronic abdominal pain, self-administering investigational retatrutide 4 mg subcutaneously once weekly
DATE	13 September 2026
SCOPE	Pharmacology · GLP-1 class evidence in IBD · immunology · nutrition · resistance training and body composition · adjunct peptides · monitoring
EVIDENCE CUT-OFF	September 2026, including TRIUMPH-1 phase 3 and the 2025 IBD meta-analyses
REVISION	Six substantive corrections to the previous draft — see §1 *Corrections*

Educational research material. Not medical advice, not a prescription, not a treatment plan.
Retatrutide is an investigational agent, not approved in any jurisdiction. BPC-157, KPV and TB-500 are unapproved research chemicals with no established human dosing. Crohn's disease carries risks of obstruction, fistula, malabsorption and surgery. Nothing here substitutes for the care of a treating gastroenterologist, and no dose in this document should be started, changed or stopped without one.

§0 Resumen ejecutivo

Este informe revisa la evidencia sobre retatrutida 4 mg semanales en una paciente con enfermedad de Crohn. La conclusión central cambió respecto al borrador anterior.

Lo nuevo. La dosis de 4 mg ya no depende de extrapolar desde el ensayo de fase 2: tiene su propio brazo en el ensayo de fase 3 TRIUMPH-1 (n = 2.339, 80 semanas), con una pérdida media de peso del **19,0 %**. Eso es eficacia sustancial y bien documentada.

La corrección importante. El borrador anterior afirmaba que los efectos gastrointestinales son «sustancialmente menores» a 4 mg. Los datos por dosis lo desmienten: la diarrea aparece en el **25,2 %** a 4 mg frente al 32,0 % a 12 mg, y el estreñimiento en el 23,8 % frente al 26,1 %. La reducción es modesta, no sustancial. En una paciente con Crohn esto significa que **una de cada cuatro semanas de diarrea puede ser del fármaco y no del brote**, y los síntomas por sí solos no permiten distinguirlos. Ese es el problema clínico que organiza todo el protocolo.

La segunda corrección. El borrador atribuía a los agonistas GLP-1 una reducción del íleo (aHR 0,58) y de las cirugías de Crohn (aHR 0,55) citando un artículo de *Frontiers in Immunology* con 167.000 pacientes. No se sostiene. La revista correcta es *Frontiers in Medicine*, las cohortes reunidas suman unos **16.242 pacientes**, la reducción de cirugía es **HR 0,61 (IC 95 % 0,44–0,84)**, y sobre el íleo **no hay asociación significativa en ninguna dirección**. El mensaje honesto es: *tranquilizador en seguridad, no demostrado en beneficio*.

Lo que realmente hay que vigilar. No es la inflamación —es la **masa magra y el estado nutricional**. Entre el 15 % y el 40 % del peso perdido con incretinas es tejido magro, y en Crohn se suma la malabsorción ileal. Las tres palancas son proteína (1,6–2,2 g/kg/día), fuerza tres veces por semana, y hierro, B12 y vitamina D medidos, no supuestos.

Sobre los péptidos añadidos. KPV y BPC-157 tienen datos únicamente en roedores; TB-500 no tiene datos humanos de eficacia de ningún tipo. Eso no los invalida como hipótesis, pero sí los sitúa varios escalones por debajo de la retatrutida en la escala de evidencia (figura 4), y conviene decirlo con esa claridad antes de añadir un tercer o cuarto compuesto no aprobado a un intestino inflamado.

> **ADVERTENCIA SOBRE EL ORIGEN DEL PRODUCTO**
>
> Todas las cifras de este informe proceden de retatrutida de grado farmacéutico administrada en ensayos clínicos. Un vial de «uso en investigación» del mercado gris no tiene contenido, pureza ni esterilidad verificados. Si la concentración real difiere de la etiquetada, la dosis administrada no es 4 mg — y entonces ninguna de las curvas dosis-respuesta de este documento describe lo que está ocurriendo. Es la limitación más grande del caso y no se resuelve con monitorización.

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HOW TO READ THE EVIDENCE TAGS

Every substantive claim carries a tag showing what kind of study stands behind it. The tag is the point: two claims can read identically in prose and rest on completely different evidence.

RCT	Randomised controlled trial in humans.
COHORT	Observational human data. Shows association; confounding by indication is rarely excluded.
GUIDELINE	Formal multi-society clinical guidance.
ANIMAL	Preclinical only. No human efficacy data exist.
NO DATA	Reasoning from mechanism alone. Treat as hypothesis.
MECHANISM	Established physiology, applied by inference to this case.

§1 Corrections to the previous draft

Six claims in the earlier version did not survive checking. Four were wrong; two were right but understated. They are listed first because two of them change what the protocol is for.

Table 1. Every substantive change from v1, with the source that forced it.

Claim in v1	What v1 said	What the evidence says	Why it matters
GI tolerability at 4 mg	"Substantially lower at 4 mg" than at 12 mg; no dose-specific figures given.	Diarrhoea 25.2% vs 32.0%; constipation 23.8% vs 26.1%; nausea 28.6% vs 42.4%. Only nausea and vomiting fall steeply. RCT	The reassurance was misplaced. Diarrhoea at this dose is the central confounder.
GLP-1 and ileus	"Fewer ileus events (aHR 0.58)".	No significant association in either direction. Event rate ≈1–2%. COHORT	Withdrawn. There is no demonstrated protection against obstruction.
GLP-1 and Crohn's surgery	"aHR 0.55", citing *Frontiers in Immunology* 2025, n ≈ 167,000.	Pooled HR 0.61 (95% CI 0.44–0.84) from 11 studies, 16,242 patients. Journal is *Frontiers in Medicine* 2025;12:1621958. COHORT	Direction holds; magnitude, precision, sample size and journal were all wrong.
Hospitalisation benefit	Implied as part of a protective class effect.	HR 0.91 (0.71–1.17) overall; 0.93 (0.78–1.10) in Crohn's. Not significant. COHORT	The class looks safe in IBD. It is not established as therapeutic.
Perioperative guidance	"2023 ASA/SAMBA multisociety statement", 24-hour liquid diet.	2024 multisociety guidance (AGA, ASMBS, ASA, ISPCOP, SAGES), 29 Oct 2024: shared decision-making, preoperative diet modification and/or rapid-sequence induction; many patients may continue therapy. GUIDELINE	Wrong year, wrong societies, and the posture shifted from "withhold" to "assess and adapt".
Efficacy at 4 mg	≈17.1% at 48 weeks, extrapolated from phase 2.	19.0% at 80 weeks in the phase 3 4 mg arm. RCT	Understated. This dose now has direct, dose-matched randomised evidence.

TWO THINGS V1 OMITTED ENTIRELY

Product provenance. Every efficacy and safety number here comes from pharmaceutical-grade drug given under trial conditions. A grey-market "research use only" vial has unverified content, purity and sterility. This is the largest uncontrolled variable in the case.

The regulatory history of BPC-157. The FDA placed it on the Category 2 bulk-substances list in 2023 over immunogenicity and impurity concerns, then removed it from that category on 15 April 2026 — a change that is *not* an authorisation to compound or use it. See §10.

One editorial change was also made: v1 carried a supplier price comparison for these compounds. A purchase-decision aid sits awkwardly inside a document whose stated purpose is education, and it has been removed. The pharmacology, the published dose ranges and the evidence grading are all retained.

§2 Retatrutide at 4 mg: pharmacology

Retatrutide (LY3437943, Eli Lilly) is a once-weekly subcutaneous single-peptide **triple agonist** at the GLP-1, GIP and glucagon receptors — the addition of glucagon agonism is what separates it from semaglutide and tirzepatide. RCT

2.1 Three arms, three different consequences

Table 2. What each receptor arm contributes, and what it implies in an inflamed gut.

Arm	Physiological effect	Consequence in Crohn's disease
GLP-1	Central hypothalamic satiety; slowed gastric emptying; the ileal brake. The dominant driver of weight loss. MECHANISM	Cuts intake in a patient who may already be malabsorbing. Slowed transit is the mechanism behind both the constipation signal and the theoretical stricture concern.
GIP	Biased signalling; augments the anorexigenic effect and appears to improve GI tolerability relative to GLP-1 alone. MECHANISM	Neutral to favourable. Likely part of why nausea, though common, is mostly mild to moderate.
Glucagon	Raises hepatic energy expenditure and lipid oxidation. Unique among approved and late-stage incretins. RCT	Adds an expenditure-side deficit on top of the intake-side deficit. In a malabsorptive gut this stacks a third negative-energy mechanism, which is why the nutritional guard-rails in §9 are not optional.
GLP-1, immune	Preclinically reduces TNF-α and IL-1β and preserves epithelial barrier integrity in colitis models. ANIMAL	The basis of the "anti-inflammatory upside" argument. It has not been demonstrated in any human IBD trial — see §4 before relying on it.

2.2 Pharmacokinetics and what 4 mg means in practice

Table 3. PK/PD properties and their practical reading for this case.

Parameter	Value	Practical implication
Half-life	≈6 days, supporting once-weekly dosing RCT	Steady state is reached roughly 4–6 weeks after a dose change. Side effects lag an escalation by 1–2 weeks, so a symptom appearing 10 days after a change is still plausibly drug-related.
Dose-proportionality	Exposure rises approximately dose-proportionally across the studied range RCT	4 mg sits at one third of the maximum studied dose — past the steep part of the efficacy curve, below the steep part of the vomiting curve.
Titration used in trials	2 mg weekly, escalating every 4 weeks to target RCT	4 mg is the lowest phase 3 maintenance dose, reached at week 4–8 of a standard schedule.
Gastric emptying	Delayed; greatest early and after each escalation MECHANISM	Alters the absorption kinetics of oral co-medication, and is the basis of the perioperative aspiration guidance in §5.
Glycaemia	Fasting glucose and HbA1c fall; the glucagon arm offsets some hypoglycaemia risk RCT	Low standalone hypoglycaemia risk. Reduced intake plus malabsorption can still produce symptomatic lows.
Regulatory status	Investigational. Not approved in any jurisdiction. FDA submission signalled for Q1 2027 GUIDELINE	No licensed supply exists. There is no approved label, no pharmacovigilance channel, and no IBD population has been studied prospectively.

§3 Efficacy and tolerability at 4 mg

−19.0%
mean weight change at 4 mg, 80 weeks (TRIUMPH-1)

25.2%
diarrhoea at 4 mg — the flare-mimicking signal

4.1%
discontinued for adverse events at 4 mg (placebo 4.9%)

2,339
participants randomised in the phase 3 trial

3.1 The dose now has its own randomised read-out

TRIUMPH-1 randomised 2,339 adults 1:1:1:1 to retatrutide 4, 9 or 12 mg or placebo for 80 weeks, starting at 2 mg weekly with escalation every 4 weeks. The cohort was severe: mean BMI 40.0 kg/m², mean baseline weight 112.7 kg. Mean weight reduction was **19.0%** at 4 mg, 25.9% at 9 mg and 28.3% at 12 mg; 45.3% of the 12 mg group lost ≥30% of body weight, and a 104-week extension in participants with baseline BMI ≥35 reached a mean of 30.3%. RCT

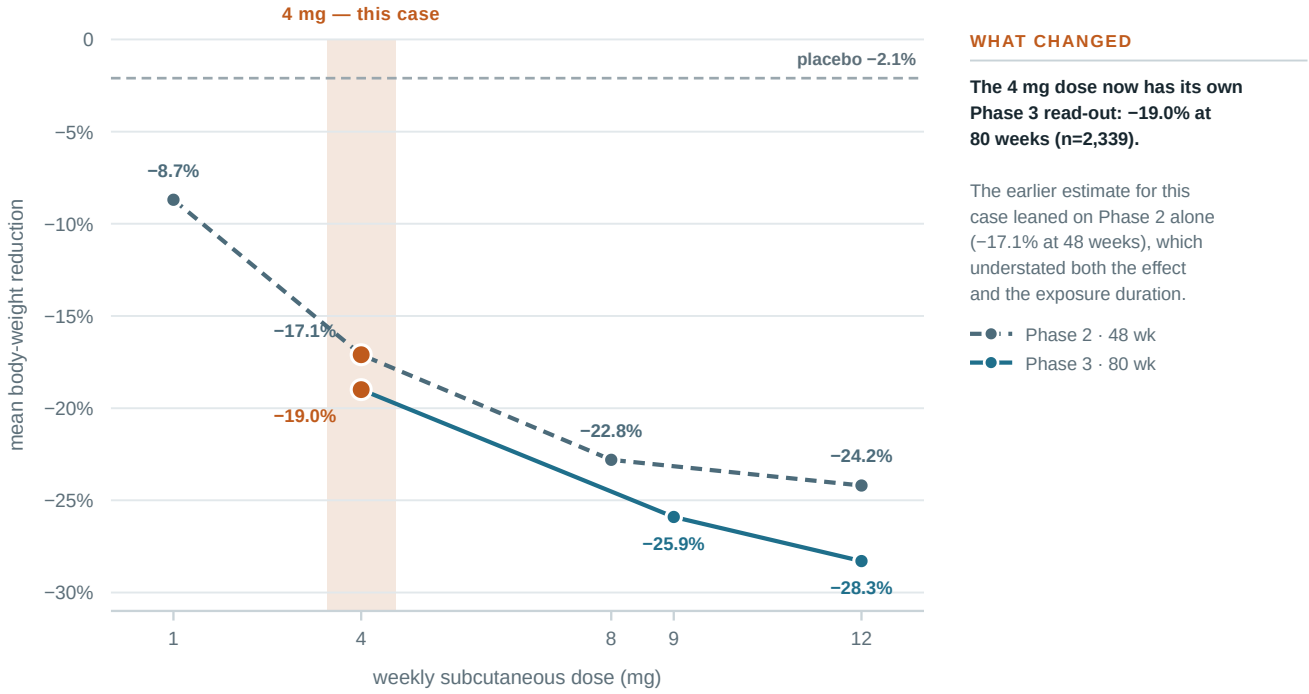


Figure 1. Dose–response for body-weight reduction. The phase 2 trial (48 weeks, n = 338) established the shape of the curve; TRIUMPH-1 (80 weeks) confirmed it at higher exposure and added a dedicated 4 mg arm. The two curves are not directly comparable — the phase 3 arm ran 32 weeks longer — but together they place 4 mg firmly on the plateau shoulder rather than the steep slope.

3.2 Tolerability: where the previous draft went wrong

Adverse events were dose-related, predominantly gastrointestinal, and mostly mild to moderate. But "dose-related" was doing a lot of unexamined work in the earlier draft. Broken out by dose, the gradient is steep for *nausea and vomiting* and shallow for *diarrhoea and constipation* — and it is diarrhoea that matters most in Crohn's disease. RCT

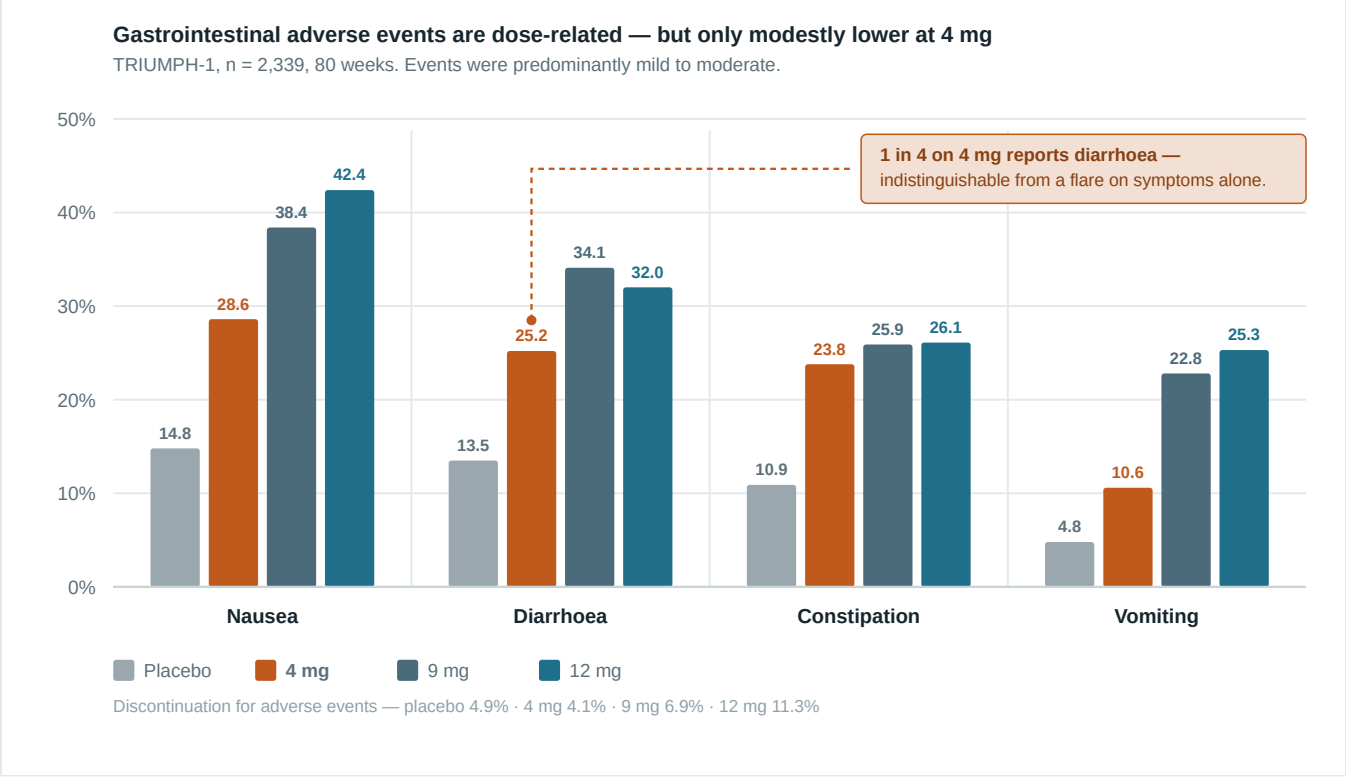


Figure 2. Gastrointestinal adverse events by dose in TRIUMPH-1. Vomiting falls by more than half from 12 mg to 4 mg (25.3% → 10.6%), and nausea by a third. Diarrhoea and constipation barely move. At 4 mg roughly one patient in four reports each of them — several times the placebo rate.

Table 4. TRIUMPH-1 adverse events by dose. Placebo column included because it is the only way to read these rates honestly: some GI symptoms occur in trials regardless of drug.

Adverse event	Placebo	4 mg	9 mg	12 mg	Reading for this case
Diarrhoea	13.5%	**25.2%**	34.1%	32.0%	The critical one. Nearly twice placebo, and only modestly below the top dose. Symptomatically identical to a flare.
Constipation	10.9%	**23.8%**	25.9%	26.1%	Essentially flat across doses. Relevant to obstructive symptoms if any stricture is present.
Nausea	14.8%	28.6%	38.4%	42.4%	Falls meaningfully at 4 mg. Manageable with timing and meal structure.
Vomiting	4.8%	10.6%	22.8%	25.3%	The clearest benefit of staying at 4 mg — less than half the 12 mg rate.
Dysaesthesia	0.9%	5.1%	12.3%	12.5%	Strongly dose-dependent. Worth knowing before any escalation is considered.
Urinary tract infection	5.3%	7.5%	8.8%	8.4%	Small excess. Note it separately if she is on a thiopurine or biologic.
Discontinued for AEs	4.9%	**4.1%**	6.9%	11.3%	At 4 mg, no worse than placebo. This dose is genuinely well tolerated in aggregate.

THE FINDING THAT REORGANISES THE PROTOCOL

At 4 mg, retatrutide is well tolerated *in aggregate* — discontinuation is no higher than placebo. But it produces diarrhoea in about a quarter of users and constipation in about a quarter. In a patient whose disease presents with exactly those symptoms, the drug installs a permanent source of diagnostic noise. Every subsequent decision in this document — the marker schedule, the seven-day rule, the algorithm in §7 — exists to cut through that noise. Symptom diaries alone cannot do it.

3.3 Body composition: the unmeasured risk

Across incretin trials, lean tissue accounts for roughly **15–40%** of total weight lost: about 25% in SURMOUNT-1 (tirzepatide) and closer to 45% in STEP 1 (semaglutide), where lean mass fell 9.7% against a 19.3% fall in fat mass. RCT No body-composition data have been published for retatrutide at these doses, and its glucagon arm makes extrapolation from the other agents less safe rather than more. NO DATA

At an expected 19% total weight loss, the plausible lean-mass loss for this patient spans roughly 3–8% of body weight — on top of the sarcopenia risk that active Crohn's disease already carries. The levers that shift the ratio back toward fat are established and few: resistance training and protein intake. Reported protein intakes in cohorts that preserved lean tissue were **1.6–2.3 g/kg fat-free mass per day** with resistance training 3–5 days per week. COHORT This is the single most actionable finding in the report and it is developed in §9.

§4 GLP-1 receptor agonists in inflammatory bowel disease

This is where the previous draft was least reliable, so the standard of proof here is deliberately strict. The honest summary: **reassuring on safety, unproven on benefit.**

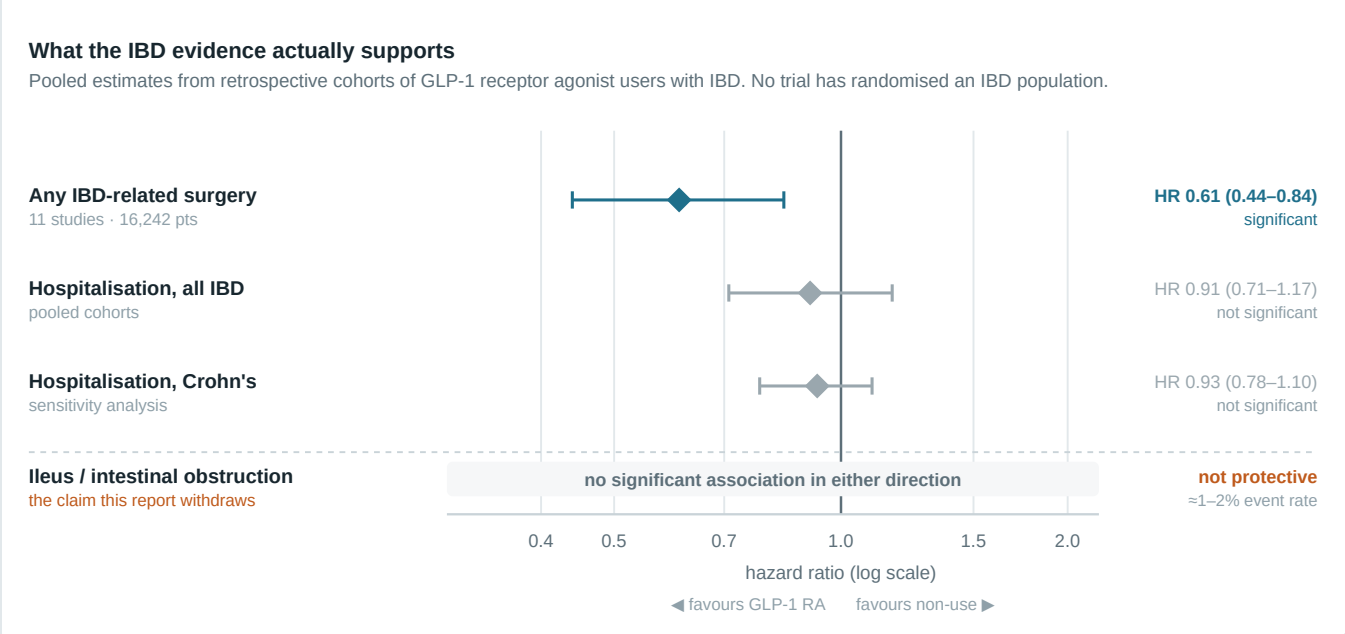


Figure 3. Pooled estimates from retrospective cohorts of GLP-1 receptor agonist users with IBD. Only the surgery endpoint reaches significance. The hospitalisation estimates cross 1.0, and no protective effect on ileus or obstruction was demonstrated — the claim v1 made and this version withdraws. No randomised trial has enrolled an IBD population.

4.1 What is actually established

- **No signal of harm.** Across multiple datasets, GLP-1 RA use was *not* associated with increased IBD exacerbations. Given how plausible a mechanistic worry it was, this is the most useful finding for a patient already taking one. COHORT
- **Surgery.** Pooled HR 0.61 (95% CI 0.44–0.84) across 11 studies and 16,242 patients — a real association, and the strongest positive signal in the literature. COHORT
- **Hospitalisation.** HR 0.91 (0.71–1.17) overall, 0.93 (0.78–1.10) in Crohn's, 0.88 (0.66–1.18) in ulcerative colitis. All cross 1.0. COHORT
- **Ileus and obstruction.** No significant association in either direction; roughly 1–2% of patients experienced such an event. COHORT
- **Steroid use.** Several large registries report reduced corticosteroid exposure among users, though pooled estimates are not consistent. COHORT

- **Weight and metabolic outcomes.** Patients with IBD and obesity achieve substantial weight loss, comparable to non-IBD populations. COHORT

4.2 Why these numbers deserve caution

All of the above is retrospective. Three limitations bear directly on how much weight to put on them:

- **Confounding by indication.** Patients prescribed a GLP-1 RA are, almost by construction, those whose disease is quiescent enough for a clinician to prioritise weight. Comparing them to non-users compares two different diseases in two different states.
- **Heterogeneity.** One meta-analysis returned an IBD-related surgery odds ratio of 0.32 with a confidence interval of 0.00 to 313.88 and $I^2 = 92.5\%$ — an interval so wide it carries no information, and a useful reminder of how unstable this literature still is.
- **No retatrutide data at all.** Every estimate above comes from semaglutide, liraglutide, dulaglutide and tirzepatide. Retatrutide's glucagon receptor agonism has no analogue in that evidence base. Extending a class effect to it is inference, not observation. NO DATA

> **HOW TO STATE THIS TO A GASTROENTEROLOGIST**
>
> "The class shows no increase in IBD exacerbations across large cohorts, and an association with reduced surgery (HR 0.61). Hospitalisation and obstruction outcomes are neutral. There is no randomised IBD evidence, and none at all for retatrutide specifically." That framing is defensible in a clinic conversation. "GLP-1s reduce Crohn's inflammation" is not.

§5 Interactions, contraindications and surgery

Table 5. Interaction and contraindication review for a Crohn's patient on an incretin.

Issue	What is known	Action
Biologics infliximab, ustekinumab, risankizumab, vedolizumab	No known pharmacokinetic interaction. These are parenteral proteins cleared by catabolism, not affected by gastric emptying. MECHANISM	No dose change indicated. Continue as prescribed; the gastroenterologist should simply know retatrutide is on board.
Thiopurines azathioprine, 6-mercaptopurine	Oral agents with narrow therapeutic margins. Delayed gastric emptying can alter absorption kinetics; no dedicated interaction study exists. NO DATA	Check FBC and LFTs 4 weeks after starting or changing the retatrutide dose, then per the standard schedule. This is the one interaction worth actively monitoring.
Any oral medication including contraception	Slowed gastric emptying shifts absorption timing; effect on total exposure is usually modest. MECHANISM	Note the theoretical risk with narrow-index drugs. Separate from injection day if symptomatic.
Stricturing disease	Delayed emptying plus the ileal brake plus a 24% constipation rate are, together, a plausible aggravator of obstructive symptoms. Not demonstrated — but the cohort data show no protection either. MECHANISM	**Prerequisite, not a recommendation:** current intestinal ultrasound or MR enterography establishing whether significant stenosis exists. Without that, dose escalation is not assessable.
Surgery and endoscopy	2024 multisociety guidance (AGA, ASMBS, ASA, ISPCOP, SAGES): shared decision-making; minimise aspiration risk through preoperative diet modification — typically clear liquids for 24 hours — and/or rapid-sequence induction. Patients without risk factors may continue therapy. GUIDELINE	Disclose the drug to every procedural and anaesthetic team, including for colonoscopy. Do not silently stop or continue.
Gallbladder disease	Rapid weight loss of any cause raises gallstone risk; incretins carry a recognised signal. RCT	Right-upper-quadrant pain warrants ultrasound rather than attribution to Crohn's.
Pancreatitis	Rare class association; causality debated. RCT	Check lipase only if clinically indicated. Routine screening is not supported.
MTC / MEN2	Rodent C-cell findings; a labelled contraindication for the approved agents in this class. ANIMAL	Personal or family history is an absolute exclusion.
Pregnancy	Contraindicated. Rapid weight loss and an unapproved agent are both disqualifying. GUIDELINE	Reliable contraception while on therapy; discontinue well before any planned conception.
Immunosuppression	Rapid weight loss and undernutrition impair immune competence in an already immunosuppressed patient. MECHANISM	Keep vaccinations current. Avoid live vaccines while on biologics or thiopurines.

§6 Baseline workup and disease mechanism

6.1 Why chronic pain in Crohn's is rarely one thing

Persistent abdominal pain in Crohn's disease usually combines three separable processes, and they call for different responses: MECHANISM

1. **Active transmural inflammation** — the Th1/Th17 axis driving TNF-α and the IL-23 pathway. Responds to disease-modifying therapy; tracks with calprotectin and CRP.

2. **Mechanical distension** from stricture or partial obstruction. Does not track with inflammatory markers, and is the process most plausibly aggravated by delayed transit.
3. **Visceral hypersensitivity and central sensitisation**, amplified by IL-1β and IL-6 acting on the CNS. Persists after mucosal healing and is frequently mistaken for treatment failure.

In a 40-year-old woman, three further contributors deserve explicit exclusion: iron and B12 deficiency from terminal ileal disease, osteopenia from cumulative corticosteroid exposure, and coexisting endometriosis, whose cyclical pain is routinely attributed to bowel disease for years.

6.2 Immunology, briefly

Crohn's disease is a dysregulated innate immune response plus an exaggerated Th1/Th17 response to commensal microbiota in a genetically susceptible host (NOD2, ATG16L1, IL23R). Hyperactivated macrophages and dendritic cells oversecrete TNF-α, IL-1β, IL-6, IL-12 and IL-23; IFN-γ and IL-17 sustain mucosal injury; a damaged epithelial barrier admits bacterial antigen to gut-associated lymphoid tissue, closing a self-sustaining loop. The IL-23/Th17 axis is the target of ustekinumab and risankizumab. MECHANISM

Faecal calprotectin remains the reference non-invasive marker of mucosal inflammation, and CRP the systemic one. Their value in this case is not diagnostic novelty — it is that they are the only instruments that can separate drug effect from disease activity when the symptoms are identical.

6.3 Baseline panel

Table 6. Required before interpreting anything else in this document.

Domain	Tests	Why
Disease activity	Faecal calprotectin, CRP, Harvey–Bradshaw Index	The reference point against which every later symptom is judged.
Structural	Intestinal ultrasound or MR enterography within 12 months	Establishes stenosis status. Gates any dose escalation.
Nutritional	FBC, ferritin + transferrin saturation, B12, folate, 25-OH-vitamin D, albumin	The domain most threatened by appetite suppression on a malabsorptive gut.
Metabolic	HbA1c, lipid panel, renal and hepatic function	Baseline for interpreting change during rapid weight loss.
Composition	Weight, waist, DXA if available, grip strength or a repeatable strength benchmark	Weight alone cannot detect the lean-mass problem described in §3.3.
Skeletal	DXA bone density	Steroid history plus chronic inflammation plus rapid weight loss converge on early osteopenia.

§7 Flare or side effect: the decision algorithm

Given a 25% background rate of drug-induced diarrhoea, the question "is this the disease or the drug?" will arise repeatedly. Answering it by impression is unreliable; the rule below makes it mechanical.

7.1 The rule, stated plainly

Any sustained change in GI symptoms lasting more than seven days triggers faecal calprotectin and CRP before it is attributed to anything. Not a diary entry, not a dose adjustment, not reassurance — a measurement. Two failure modes this prevents, in both directions:

- **Attributing a flare to the drug** delays escalation of disease-modifying therapy while transmural inflammation continues. This is the more dangerous error.
- **Attributing a drug effect to a flare** triggers unnecessary steroids, imaging and anxiety, and may cost her a therapy that is working metabolically.

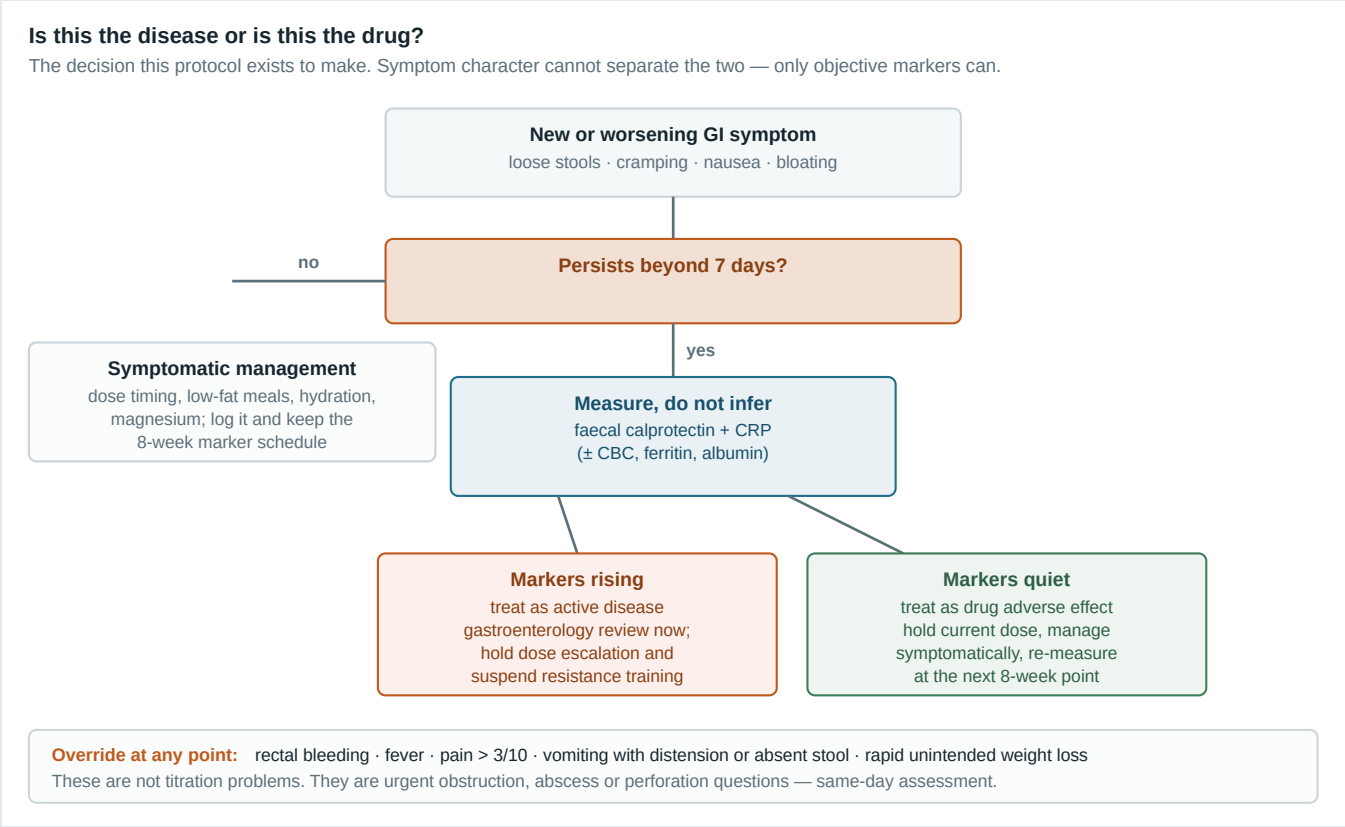


Figure 5. The seven-day rule. Transient GI symptoms after a dose or a dietary change are managed symptomatically. Anything persisting beyond a week is measured, not interpreted. Red-flag features bypass the algorithm entirely.

7.2 Features that discriminate before the markers return

Table 7. Clinical features with some discriminatory value. Suggestive only — they do not replace measurement.

Feature	Favours drug effect	Favours active disease
Timing	Begins within 1–2 weeks of a dose change; worst in the days after injection	No relationship to injection timing or dose history
Trajectory	Attenuates over 2–4 weeks at a stable dose	Static or progressive
Blood in stool	Absent	Present — investigate regardless of markers
Nocturnal symptoms	Uncommon	Waking from sleep to defecate is a classic organic feature
Systemic features	Absent	Fever, night sweats, arthralgia, erythema nodosum
Nausea pattern	Prominent, meal-related, improves with smaller low-fat meals	Nausea with distension and absent flatus suggests obstruction, not intolerance
Weight	Loss is gradual and expected on the treatment curve	Loss accelerates away from the expected curve

RED FLAGS — SAME-DAY ASSESSMENT, NOT NEXT-CYCLE REVIEW

Rectal bleeding · fever · abdominal pain above 3/10 or unremitting · vomiting with distension or absent stool and flatus · rapid unintended weight loss beyond the expected curve · a palpable tender mass. These are obstruction, abscess and perforation questions. They are not titration problems, and no monitoring schedule in this document should delay acting on them.

§8 Nutrition protocol

Built on the Crohn's Disease Exclusion Diet, which has randomised evidence for inducing remission, RCT and then adapted for the specific problem that appetite is now pharmacologically suppressed.

Table 8. Twelve-week structure. Phase boundaries are governed by symptoms and markers, not the calendar.

Phase	Weeks	Included	Excluded	Objective
1 · Calm-down	1–6	White rice, boiled potato, eggs, chicken and turkey, white fish, cooked carrot and squash, banana, some white bread, olive oil, strained broth; fermented dairy if tolerated (unsweetened kefir, Greek yoghurt)	Gluten, most dairy, added sugar, ultra-processed foods, alcohol, whole nuts and seeds, legumes, raw insoluble fibre, polyol sweeteners	Mucosal rest; reduce fermentable substrate; establish a clean symptom baseline
2 · Reintroduction	7–12	Adds gluten-free oats, sweet potato, cooked courgette, cooked apple and berries, hard aged cheeses, oily fish twice weekly, cooked turmeric and ginger	Continues to exclude gluten, added sugar, ultra-processed foods, alcohol, polyols	Restore microbial diversity without provoking symptoms
3 · Maintenance	13+	Soft Mediterranean pattern; one new food every 3–4 days with a symptom log (pain 0–10, stool form and frequency, bloating)	Only foods that reproducibly provoke symptoms on two separate challenges	Sustainability and nutrient density — which is now the binding constraint

8.1 Rules that change because of the drug

- **Protein first at every meal.** Satiety arrives early and unpredictably. Whatever is eaten first is what actually gets eaten — so it should be the protein.
- **Five to six small meals.** Large meals worsen Crohn's pain and incretin nausea by the same mechanism. This is the rare instance where both problems have one answer.
- **Liquid protein as a floor, not a treat.** On low-appetite days a shake is often the only realistic route to the target. Plan for it rather than treating it as failure.
- **Everything cooked in phase 1**, chewed thoroughly, with 2–2.5 L of fluid daily — which also addresses the 24% constipation rate.
- **Avoid NSAIDs.** Unchanged, and independent of the drug.

8.2 Supplements

Table 9. Graded by evidence, not by popularity.

Agent	Typical dose	Rationale and standing
Protein (whole food or supplement)	1.6–2.2 g/kg/day	The non-negotiable one. Directly counters the lean-mass loss described in §3.3. RCT
Vitamin D3	2,000–4,000 IU/day, titrated to 25-OH-D	Deficiency is near-universal in Crohn's and relevant to both bone and immune function. COHORT
Iron	Per ferritin and transferrin saturation	Oral iron is frequently poorly tolerated in active disease; intravenous replacement is often preferred. A clinician decision. GUIDELINE
Vitamin B12	Per level; parenteral if ileal disease or resection	Terminal ileal absorption is the exact site at risk. GUIDELINE
Omega-3 (EPA/DHA)	2–3 g/day	Reasonable for general cardiometabolic benefit; trials for maintaining Crohn's remission were negative. RCT
Creatine monohydrate	3–5 g/day	Strong general evidence for lean-mass preservation during energy deficit; not studied in IBD. RCT
Curcumin	500–1,000 mg, bioavailable form	Some randomised support as an adjunct in ulcerative colitis; evidence in Crohn's is weaker. RCT
L-glutamine	5 g/day	Popular on barrier-function reasoning; human trials in Crohn's have not shown benefit. Low priority. NO DATA
Multivitamin	Daily	Justified specifically because total intake is now pharmacologically reduced. MECHANISM

§9 Training and body composition

Moderate exercise is anti-inflammatory through muscle-derived IL-6 and regulatory T-cell modulation; high-volume, high-intensity work raises TNF-α and IL-1β and is a plausible flare trigger. COHORT On a triple agonist, resistance training moves from beneficial to **load-bearing**: it is one of only two levers protecting lean tissue.

Table 10. Programme parameters, and what each is protecting against.

Element	Prescription	Rationale
Frequency	Resistance training 3×/week on non-consecutive days; zone-2 walking 20–40 min on alternate days; 1–2 full rest days	Three sessions is the documented threshold for lean-mass preservation during incretin therapy.
Intensity	RPE 4–6/10; roughly 60–75% 1RM; 8–15 reps; 2–4 sets. Never to failure during active disease	Keeps training in the anti-inflammatory band and off the flare-triggering end.
Session	Goblet squat or leg press → single-arm row → bench or cable press → light hinge (RDL or glute bridge) → anti-extension core (dead bug, Pallof press) → 5 min diaphragmatic breathing	Compound movements across the largest muscle mass; no loaded spinal flexion; no Valsalva-heavy work.
Protein	1.6–2.2 g/kg body weight daily, or 1.6–2.3 g/kg fat-free mass	Without this, the training does not protect lean tissue. The two work only together.
Energy intake	Maintenance, or a deficit no greater than 10%	**Do not diet aggressively on this drug.** Retatrutide supplies the deficit. Stacking a voluntary one accelerates lean loss without adding fat loss.
Progression	Add one set or 2.5 kg only when HBI ≤4 and no residual pain at 48 h. Deload 40% of volume every fourth week	Progression gated on disease state rather than the calendar.
During a flare	10–20 min easy walking, mobility and breathing only. No loading until pain ≤1/10 and calprotectin is falling	Training through a flare is the fastest route to losing both the training block and the remission.
Recovery	7–9 h sleep; deliberate stress management	Short sleep raises IL-6 and TNF-α; psychological stress is among the better-documented relapse triggers. COHORT

MEASURE THE RIGHT THING

Scale weight will fall on this drug regardless of what the training does — which makes it useless as feedback. Track instead: waist circumference, a repeatable strength benchmark (a fixed load for maximum reps), grip strength, and DXA lean mass if accessible. If strength is holding while weight falls, the protocol is working. If strength falls with weight, protein and training are losing and the plan needs to change, not the dose.

§10 Adjunct peptides: evidence and regulatory standing

The compounds proposed as adjuncts differ enormously in how much is actually known about them. Presenting them in one table implies an equivalence that does not exist, so the evidence tier is shown first.

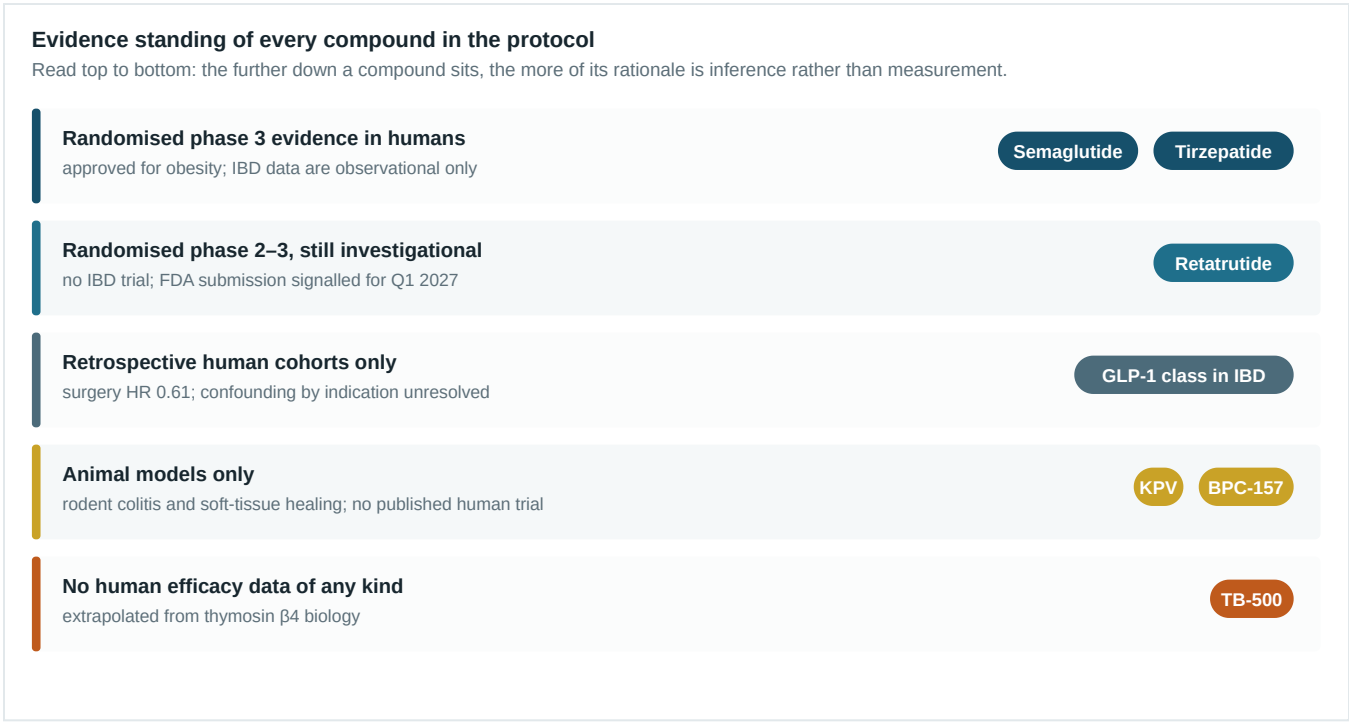


Figure 4. Evidence tier for every compound under discussion. Retatrutide sits on randomised phase 3 data — for obesity, not for Crohn's. KPV and BPC-157 rest on rodent models. TB-500 has no human efficacy data of any kind.

Table 11. Mechanism, published research ranges and standing. Ranges are transcribed from the research literature for documentation. They are not recommendations and no dose here has an established human safety margin.

Compound	Proposed mechanism	Ranges in the literature	Standing and cautions
KPV α-MSH fragment	Melanocortin-pathway anti-inflammatory; reduces TNF-α and IL-1β and protects the epithelial barrier in murine colitis ANIMAL	200–500 µg/day subcutaneous in reported experimental use; 4–8 week blocks	Mechanistically the best fit for intestinal inflammation of anything in this table. Evidence is entirely preclinical — the Kannengiesser 2008 murine work remains the principal citation. No human trial.
BPC-157	Gastric and intestinal cytoprotection, angiogenesis, accelerated healing in rodent colitis and soft-tissue models ANIMAL	250–500 µg once or twice daily; 4–12 weeks	Placed on the FDA Category 2 bulk-substances list in 2023 over immunogenicity, peptide-related impurities and inadequate API characterisation; removed from that category on 15 April 2026, which is **not** an authorisation to compound or use. No published human trial has reported results. Prohibited in sport (USADA/WADA).
TB-500 thymosin β4 fragment	Actin sequestration, cell migration, tissue repair NO DATA	2–2.5 mg twice weekly for 4–6 weeks, then 1–2 mg weekly	The weakest evidence position of any compound here. Systemic exposure with no human efficacy data. Prohibited in sport.
Retatrutide current therapy	GLP-1 + GIP + glucagon triple agonist RCT	Trial titration: 2 mg weekly, +2 mg every 4 weeks to 4, 9 or 12 mg	Robust randomised evidence in obesity; none in IBD. Investigational — FDA submission signalled for Q1 2027. **Hold 4 mg for at least 8–12 weeks and review markers before considering any escalation.**
Semaglutide **Tirzepatide**	GLP-1, and GLP-1 + GIP respectively RCT	Semaglutide 0.25 → 2.4 mg weekly; tirzepatide 2.5 → 15 mg weekly	Approved, prescribable, pharmacovigilance in place, and the agents that generated the IBD cohort data in §4. If regulated supply and clinical oversight matter more than maximum efficacy, this is the rational alternative.

10.1 A defensible sequence

Adding two or three unapproved compounds simultaneously to an inflamed gut makes attribution impossible: if something improves or worsens, nothing will indicate which agent did it. If adjuncts are pursued at all, sequence them and measure between steps.

1. **Weeks 0–8 — change nothing pharmacological.** Diet phase 1, the protein floor, three weekly training sessions, sleep. Hold retatrutide at 4 mg. Measure calprotectin and CRP at week 8. Most of the achievable benefit lives in this step, and it is the only one with no added risk.
2. **If markers remain elevated** and the gastroenterologist agrees: introduce a single agent — KPV has the most relevant preclinical rationale — for 6–8 weeks, with markers before and after.
3. **Only then**, and only if weight and body-composition goals justify it while GI symptoms are controlled and stenosis excluded, consider discussing escalation from 4 mg.

Stop any adjunct immediately for pain above 3/10, bleeding, fever, injection-site infection or rapid unintended weight loss.

> **THE HONEST POSITION ON THE ADJUNCTS**
>
> KPV, BPC-157 and TB-500 are unapproved research chemicals. The relevant risk is not only that they may not work — it is that grey-market material carries unverified identity, purity, endotoxin load and sterility, injected by a patient who is immunosuppressed. That combination is the specific hazard, and it is not mitigated by careful dosing or by monitoring. It can only be mitigated by not doing it, or by doing it under clinical supervision with material of verified provenance.

§11 Monitoring schedule

Table 12. Rolling eight-week cycle at a stable 4 mg dose. The cycle restarts on any dose change.

Week	Actions	Decision it informs
0	Full baseline panel (Table 6). HBI. Weight, waist, photographs, strength benchmark. Confirm stenosis status. Begin diet phase 1. Hold retatrutide dose constant.	Establishes the reference point. Without it, nothing later is interpretable.
2	Review GI symptoms against Table 7. Audit actual protein intake for three days — measured, not estimated. Confirm training is stable at RPE 4–6.	Early detection of an intake shortfall, which is the most common and most correctable failure.
4	FBC and LFTs if on a thiopurine. Weight and waist. Formal flare-versus-adverse-effect review.	Catches absorption-related haematological drift and consolidates the symptom picture.
8	Faecal calprotectin, CRP, ferritin and transferrin saturation, B12, folate, 25-OH-D, HbA1c, lipids, albumin. Weight, waist, strength, pain score, sleep quality.	**The decision point.** Markers quiet and strength holding → continue unchanged. Markers rising → gastroenterology, not adjuncts. Strength falling → fix protein and training before touching the dose.

> **INTERPRETING THE WEEK-8 PANEL**
>
> The panel answers three questions in order of priority. **Is the disease active?** Calprotectin and CRP. **Is she becoming depleted?** Ferritin, B12, folate, vitamin D, albumin. **Is she losing the right tissue?** Strength benchmark against weight change. A good result on weight with a bad result on any of the other three is not a good result.

§12 Ten priorities, ranked

Table 13. Ordered by consequence if neglected, not by how interesting they are.

#	Priority	Why it ranks here	Concrete action
1	**Separating flare from side effect**	A 25% drug-induced diarrhoea rate means symptoms have permanently lost their diagnostic value. Getting this wrong in either direction is the highest-consequence error available.	The seven-day rule (§7). Calprotectin and CRP before attribution, every time.
2	**Defending lean mass**	15–40% of the ~19% expected weight loss can be lean tissue, on top of existing sarcopenia risk. Once lost it is slow and difficult to regain.	Protein 1.6–2.2 g/kg/day and three weekly resistance sessions. Track strength, not weight.
3	**Product provenance**	If the vial does not contain what the label says, every number in this report describes a different drug than the one being injected.	Verified supply and clinical supervision, or an approved alternative (§10).
4	**Stenosis status**	Determines whether delayed transit is a nuisance or a hazard, and gates any dose escalation.	Current intestinal ultrasound or MR enterography before any change.
5	**Micronutrient depletion**	Suppressed intake plus ileal malabsorption compound. Fatigue from anaemia is routinely misread as disease activity.	Ferritin >50 ng/mL, plus B12, folate and vitamin D repleted and re-measured at eight weeks.
6	**Gastroenterology co-management**	An unapproved triple agonist alongside biologics or thiopurines is not a self-managed situation.	Disclose the drug and dose. Ask specifically about stenosis and thiopurine monitoring.
7	**Training dose**	Moderate loading is anti-inflammatory; excessive loading is a flare trigger. The window is real and narrow.	RPE 4–6, deload every fourth week, stop if pain exceeds 2/10.
8	**Sleep and stress**	Short sleep raises IL-6 and TNF-α; stress is among the best-documented relapse triggers. Cheap to fix, easy to ignore.	7–9 h sleep; ten minutes daily of breathing work or a CBT-based practice.
9	**Bone health**	Steroid history, chronic inflammation and rapid weight loss all reduce bone density, silently, at 40.	DXA. Load-bearing resistance work. Vitamin D, K2 and calcium adequacy.
10	**Adjunct peptide sequencing**	Ranks last deliberately: the weakest evidence, and adding several at once destroys attribution.	One agent at a time, markers either side, only after the eight-week foundation block.

§13 Appendix: deep-research prompt

For re-running this review against newer evidence. Updated with the corrected figures so a future model is not re-anchored on the errors in v1.

```
You are a multidisciplinary team: an IBD gastroenterologist, an immunologist, a CDED-certified
clinical dietitian, an exercise physiologist and a peptide pharmacologist. Produce a rigorously
cited report (PubMed, NEJM, Lancet, Gut, J Crohns Colitis; last 5 years) on a case: a
40-year-old woman with ileal Crohn's disease and chronic abdominal pain, self-administering
INVESTIGATIONAL RETATRUTIDE 4 mg subcutaneously once weekly.

Verify each of the following before using it; correct anything that does not hold up, and say so.

1. PHARMACOLOGY FIRST. Retatrutide (LY3437943): GLP-1 + GIP + glucagon triple agonism with
   biased GIP signalling; half-life approximately 6 days; approximately dose-proportional
   exposure. Efficacy: phase 2 (Jastreboff, NEJM 2023;389:514-526), 48-week least-squares mean
   weight change -8.7% (1 mg), -17.1% (4 mg combined), -22.8% (8 mg combined), -24.2% (12 mg),
   -2.1% placebo. Phase 3 TRIUMPH-1: n=2,339, 1:1:1:1, 80 weeks, 2 mg start with 4-weekly
   escalation, mean baseline BMI 40.0 kg/m2 and weight 112.7 kg; -19.0% (4 mg), -25.9% (9 mg),
   -28.3% (12 mg); 45.3% achieved >=30% at 12 mg; up to -30.3% at 104 weeks in a BMI>=35
   extension.
```

```
2. DOSE-STRATIFIED TOLERABILITY, which is the clinical crux. TRIUMPH-1, 4 / 9 / 12 mg vs
   placebo: nausea 28.6 / 38.4 / 42.4 vs 14.8%; diarrhoea 25.2 / 34.1 / 32.0 vs 13.5%;
   constipation 23.8 / 25.9 / 26.1 vs 10.9%; vomiting 10.6 / 22.8 / 25.3 vs 4.8%; dysaesthesia
   5.1 / 12.3 / 12.5 vs 0.9%; discontinuation for AEs 4.1 / 6.9 / 11.3 vs 4.9%. Analyse
   explicitly what a ~25% drug-induced diarrhoea rate does to symptom-based assessment of
   Crohn's disease activity.

3. GLP-1 CLASS EVIDENCE IN IBD, stated without overreach. Frontiers in Medicine
   2025;12:1621958; J Crohns Colitis 2025;19(10):jjaf181, 19(9):jjaf167, 19(12):jjaf193.
   Surgery pooled HR 0.61 (0.44-0.84), 11 studies, 16,242 patients; hospitalisation HR 0.91
   (0.71-1.17), Crohn's 0.93 (0.78-1.10), UC 0.88 (0.66-1.18); no significant association with
   ileus or intestinal obstruction; no increase in IBD exacerbations. State the confounding-by-
   indication problem and that no RCT has enrolled an IBD population, and none exists for
   retatrutide.

4. INTERACTIONS AND PERIOPERATIVE CARE. Biologics, thiopurines (delayed gastric emptying and
   oral absorption kinetics), stricturing disease. 2024 multisociety perioperative guidance
   (AGA, ASMBS, ASA, ISPCOP, SAGES, 29 Oct 2024).

5. BODY COMPOSITION. Lean mass as a fraction of incretin-induced weight loss (~25% in
   SURMOUNT-1, ~45% in STEP 1, ~15-40% across reviews); note the absence of retatrutide
   body-composition data and why its glucagon arm makes extrapolation unsafe. Protein targets
   1.6-2.3 g/kg fat-free mass; resistance training 3-5 days/week.

6. NUTRITION. CDED protocol (Levine, Gastroenterology 2019;157:440-450) adapted for
   pharmacological appetite suppression: protein-first meals, 5-6 small meals, liquid protein
   as a floor. Grade every supplement by evidence level.

7. ADJUNCT PEPTIDES. KPV (Kannengiesser, Inflamm Bowel Dis 2008;14:324-331 - murine only),
   BPC-157 (rodent only; FDA Category 2 bulks list 2023, removed 15 April 2026, which is not an
   authorisation), TB-500 (no human efficacy data). Give mechanism, published research ranges,
   interaction considerations with biologics and thiopurines, and regulatory status. Address
   grey-market provenance risk in an immunosuppressed patient.

8. DELIVERABLES. A flare-versus-adverse-effect decision algorithm; an 8-week monitoring
   schedule; a ranked priority matrix; explicit red flags. Tag every claim with its evidence
   level (RCT / cohort / animal / no human data / guideline / mechanism).

FORMAT: self-contained HTML, vector SVG figures, numbered bibliography with DOI or PMID,
>= 2,500 words. Flag every claim you could not verify rather than reproducing it.
```

§14 References

Items marked **[verified]** were checked against the source or an indexed abstract while preparing this revision. Items marked **[secondary]** are cited from reporting of the primary source, which was not directly accessible.

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4. The role of GLP-1 receptor agonists in IBD-related surgery and IBD-related complications among patients with metabolic comorbidities: a systematic review and meta-analysis. *Frontiers in Medicine* 2025;12:1621958. PMID 40917833. **[verified]**
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6. Can GLP-1 agonists be used safely in inflammatory bowel disease? A meta-analysis. *J Crohns Colitis* 2025;19(12):jjaf193. **[verified]**
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9. Torres J, Mehandru S, Colombel J-F, Peyrin-Biroulet L. Crohn's disease. *Lancet* 2017;389:1741–1755. **[verified]**

10. Levine A, Wine E, Assa A, et al. Crohn's disease exclusion diet plus partial enteral nutrition induces sustained remission in a randomized controlled trial. *Gastroenterology* 2019;157(2):440–450. **[verified]**

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12. Gwyer D, Wragg NM, Wilson SL. Gastric pentadecapeptide body protection compound BPC 157 and its role in accelerating musculoskeletal soft tissue healing. *Cell Tissue Res* 2019;377:153–159. **[secondary]**

13. US Food and Drug Administration. Certain bulk drug substances for use in compounding that may present significant safety risks (503A Category 2 list); BPC-157 listed 2023, removed from Category 2 on 15 April 2026. **[verified]**

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CLOSING STATEMENT

This document is educational research material prepared at the request of the person described. It is not medical advice and creates no clinical relationship. Retatrutide is investigational and not approved in any jurisdiction; KPV, BPC-157 and TB-500 are unapproved research chemicals. Doses reproduced here are transcribed from published literature for documentation and are not recommendations. Crohn's disease is a serious condition with risks of obstruction, fistula, malabsorption and surgery, and its management — including any decision about a GLP-1-class agent — belongs with a treating gastroenterologist.

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